Aaron Dutta

647-244-5263 | aaron.dutta@uwaterloo.ca | LinkedIn | GitHub | aarondutta.com

Education

University of Waterloo Sep 2026 - Apr 2031
BMath, Honours Mathematics with Co-Op *Waterloo, ON*

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL
Frameworks: React, Next.js, Node.js, FastAPI, Flask, Hono, Prisma, REST APIs
AI/ML and Data: OpenCV, MediaPipe, NumPy, Pandas, Matplotlib, Pydantic, LangChain
Tools and Cloud: Git, Docker, PostgreSQL, Redis, Azure, GCP

Experience

Waterloo Aerial Robotics Group Oct 2026 – Present
Autonomy Software Engineer *Waterloo, ON*

- Extended the team's single-drone survey planner to cover one mapping area with **any number of drones**, splitting it into non-overlapping routes that share altitude and photo overlap
- Tested in simulation that the combined photo set matches a single-drone survey, and that one drone reproduces the original plan

Magic Hour AI (YC W24) Sep 2026 – Present
Software Engineering Intern *Remote / San Francisco, CA*

- Launched 3 live apps on Magic Hour's video-generation API to grow adoption of the public developer API; launch posts reached **1,000+ views on X**
- Wrote the backends in Hono, PostgreSQL, and Redis for a group-chat awards video app
- Reduced each 100k-message chat to a **20k-token summary** for an AI decision model (JEV) to judge, so it fits the model's window at about **$0.008** per judgment

UW Cube Aug 2026 – Present
Design Engineer *Waterloo, ON*

- Developing Cube's public website (Next.js, TypeScript) and integrating its AI chatbot (Python, LangChain backend services), which helps visitors find information and navigate the site
- Reviewing pull requests from software engineers on a 16-person product team to keep design and code aligned

Projects

TechniqueTitan | *Python, MediaPipe, OpenCV, NumPy, FastAPI, Pandas, Matplotlib* | Jul 2026 - Present

- Created a piano posture coach that gives hand-form feedback in seconds from any device camera, with **10 users** already using the product during development
- Designed a computer-vision pipeline that scores 5 posture criteria from 21 hand landmarks, with a normalization stage to reduce dependence on camera distance
- Automated analysis of a **115-image** labeled dataset with a batch tool that exports CSV/JSON metrics, replacing my image-by-image review

SkyEye | *Python, Flask, YOLOv8, ONNX Runtime, React, TypeScript* | Sep 2026

- Developed a search-and-rescue tool that turns a free-text missing-person report into a search area and ranks person-candidate pins by detector confidence in **5–8 seconds**
- Tuned tile size on a sliding-window YOLOv8n detector (ONNX Runtime), raising person recall from **4/10 to 7/10**
- Implemented a Flask and React operator console using Gemini Flash (Groq fallback) to extract report details, and GCP to geocode locations and draw search zones

Awards

- **Winner** @ RescueHacks, 160+ hackers Sep 2026
- **2nd Place** Overall @ Katy Youth Hacks, 220+ hackers Aug 2026
- **3rd Place** Security/Privacy Hack @ STEMist Hacks, 330+ hackers Aug 2026
- **1st Place** @ STEMMINDS Hackathon, 60+ hackers Mar 2026